

Programming Exercises in C (Sequential Programming)

Easy Problems

1. Cumulative Sum

Compute the sum $1 + 2 + \dots + n$.

Task: Write a C program that takes n as input and computes the sum using sequential operations.

The formula for summation is: $n(n+1)/2$

Example:

Input: $n = 5$

Output: 15

2. Coin Breakdown (Centavos)

Break down a given centavo amount into coins: 25, 10, 5, 1.

Task: Write a C program (sequential only) to output the number of coins for each denomination.

Example:

Input: 87 centavos

Output: 3:25, 1:10, 0:5, 2:1

3. Temperature Conversion

Convert Celsius input to Fahrenheit and Kelvin.

· - $F = (9/5)C + 32$

· - $K = C + 273.15$

Task: Write a C program that asks for Celsius and outputs Fahrenheit and Kelvin.

Example:

Input: $C = 25$

Output: $F = 77$, $K = 298.15$

Medium Problems

1. Effective Rate of Interest

The formula is: $r_{\text{eff}} = (1 + r/m)^m - 1$

- - r = nominal interest rate
- - m = number of conversion periods per year

Task: Write a C program that computes the effective interest rate.

Example:

Input: $r = 0.08$, $m = 4$

Output: $r_{\text{eff}} = 0.082432$

2. Peso Breakdown

Break down a peso amount into denominations: 1000, 500, 100, 50, 20, 10, 5, 1.

Task: Write a C program (sequential only) that outputs the breakdown.

Example:

Input: 1538 pesos

Output: 1:1000, 1:500, 0:100, 0:50, 1:20, 1:10, 1:5, 3:1

3. Distance Conversion

Convert a distance in meters into kilometers, centimeters, and millimeters.

Task: Write a C program to perform the conversion.

Example:

Input: 1234 meters

Output: 1.234 km, 123400 cm, 1234000 mm

Hard Problems

1. Halley's Law (Barometric Pressure)

$p(x) = 29.29 * e^{(-0.2x)}$, where e is Euler's number.

- - Input: altitude x in miles
- - Compute $p(x)$ using `exp()`
- - Output pressure in inHg and mmHg (1 in = 25.4 mm)

Task: Write a C program that performs this computation.

Example:

Input: $x = 3$ miles

Output: $p(x) = 17.99$ inHg, 456.9 mmHg

2. Consumer Price Index (CPI)

$$I(t) = -0.2t^3 + 3t^2 + 100$$

Task: Write a C program that asks for t (years) and computes $I(t)$.

Example:

Input: $t = 5$

Output: $I(t) = 135$

3. Time Conversion (Seconds → H:M:S)

Convert seconds into hours, minutes, and seconds.

Task: Write a C program (sequential only) to perform this conversion.

Example:

Input: 3875 seconds

Output: 1 hour, 4 minutes, 35 seconds

4. Radioactive Decay

$$N(t) = N_0 * e^{(-\lambda t)}$$

- - N_0 = initial quantity
- - λ = decay constant
- - t = time

Task: Write a C program that asks for N_0 , λ , t and computes $N(t)$ using $\exp()$.

Example:

Input: $N_0 = 100$, $\lambda = 0.05$, $t = 10$

Output: $N(t) = 60.65$

5. Population Growth

$$P(t) = P_0 * (1 + r)^t$$

- - P_0 = initial population
- - r = growth rate
- - t = years

Task: Write a C program that computes population growth.

Example:

Input: $P_0 = 1000$, $r = 0.02$, $t = 10$

Output: $P(t) = 1219$

6. Time Breakdown (Days → Years/Months/Days)

Convert a given number of days into years (365 days), months (30 days), and days.

Task: Write a C program to perform this conversion sequentially.

Example:

Input: 800 days

Output: 2 years, 2 months, 10 days

7. Turuan mo ako paano magmahal :(

Pinakalisod sa tanan